

Real vs. Fake Face Detection — Comprehensive Results Report

Master 2 Thesis — Deepfake Detection Track — Phases 1-10 Complete

1. Objective

Build and validate an end-to-end pipeline for digital face forgery (deepfake) detection, then progressively evaluate and improve it. This report consolidates all ten phases of the experimental work: the four items originally requested by the supervisor (fine-tuning, cross-dataset evaluation, ensembling, adversarial robustness), plus six self-directed extensions investigating the fixability, extent, representation-dependence, few-shot recoverability, and backbone-dependence of the generalization gap found.

2. Dataset & Setup

- Primary dataset (Phases 1, 2, 4, 5, 6, 8, 9): "140k Real and Fake Faces" (Kaggle) — real photographs (FFHQ) vs. StyleGAN-generated faces.
- Cross-domain dataset (Phases 3, 6, 8, 9): "Deepfake and Real Images" (Kaggle) — real face-swap deepfake video frames.
- Third-domain dataset (Phases 7, 8, 9): "Face Dataset Using Stable Diffusion v.1.4" (Kaggle) — Stable Diffusion-generated faces.
- Base architectures: ResNet18 and EfficientNet-B0, both pretrained on ImageNet. Training: Colab T4 GPU, Adam optimizer, batch size 64.

3. Phase 1 — Baseline (frozen backbone, linear probe)

- Only the final classification layer trained (1,026 parameters); entire ResNet18 backbone frozen at ImageNet weights.
- Result: 80.8% test accuracy, 0.888 AUC.

Confusion matrix (n = 20,000):

	Pred: Fake	Pred: Real
True: Fake	8050	1950
True: Real	1896	8104

4. Phase 2 — Deeper Fine-Tuning (unfreeze layer4)

Unfreezing the last residual block (layer4) of ResNet18 allowed the network's highest-level convolutional features to adapt directly to the task, while earlier layers remained frozen.

- 8,394,754 trainable parameters (vs. 1,026 in Phase 1).
- Result: 98.8% test accuracy, 0.999 AUC.

Confusion matrix (n = 20,000):

	Pred: Fake	Pred: Real
True: Fake	9909	91
True: Real	146	9854

5. Phase 1 vs. Phase 2 — Comparison

Metric	Baseline #1 (frozen backbone)	Phase 2 (fine-tuned layer4)
Trainable parameters	1,026	8,394,754
Test Accuracy	80.8%	98.8%
Test AUC	0.888	0.999
Precision / Recall / F1 (avg)	0.81 / 0.81 / 0.81	0.99 / 0.99 / 0.99

6. Phase 3 — Cross-Dataset Evaluation (Generalization / Drift)

Both trained models were evaluated, without any retraining, on an independent dataset built from real face-swap deepfake videos. Test set size: 10,905 images.

Model	In-dist. Acc / AUC	Cross-dataset Acc / AUC
Baseline #1 (frozen backbone)	0.808 / 0.888	0.487 / 0.489
Phase 2 (fine-tuned layer4)	0.988 / 0.999	0.495 / 0.525

Interpretation

- Both models collapse to near-chance performance on the unseen dataset (AUC 0.489 and 0.525, vs. 0.888 and 0.999 in-distribution).
- Both models are heavily biased toward predicting "Real" on this unseen data rather than failing randomly.

7. Phase 4 — Ensemble (ResNet18 + EfficientNet-B0)

A second, independently trained architecture (EfficientNet-B0) was combined with the Phase 2 ResNet18 via simple probability averaging.

Model	In-dist Acc	In-dist AUC	Cross Acc	Cross AUC
ResNet18 (Phase 2)	98.81%	0.999	49.45%	0.525
EfficientNet-B0	93.82%	0.987	48.88%	0.499
Ensemble (avg)	98.99%	0.999	49.35%	0.498

Interpretation

- On the cross-dataset test, the ensemble's AUC (0.498) is slightly lower than ResNet18 alone (0.525) — architectural diversity alone does not resolve the generalization gap, since both models share the same single-source blind spot.

8. Phase 5 — Adversarial Robustness

The Phase 2 model was evaluated against FGSM (single-step) and PGD (10-step iterative) white-box adversarial attacks across a range of perturbation strengths (epsilon).

Epsilon	FGSM Accuracy	PGD Accuracy
0.00 (clean, no attack)	99.0%	99.0%
0.01	0.0%	0.0%
0.03	0.1%	0.0%
0.05	13.2%	0.0%
0.10	44.9%	0.0%

Interpretation

- At epsilon = 0.01 — a near-imperceptible perturbation — accuracy collapses to 0% under both attacks. PGD remains at 0% across all tested epsilon values.
- FGSM's non-monotonic behavior at larger epsilon is a documented property of its single-step linear approximation breaking down, not a recovery of robustness.

9. Phase 6 — Mixed-Source Training

A fresh ResNet18 was fine-tuned (identical recipe to Phase 2) on a combined training set drawn from both the StyleGAN and face-swap sources.

Model	Test Set	Accuracy	AUC
Phase 2 (single-source)	StyleGAN	98.68%	0.999

Model	Test Set	Accuracy	AUC
Phase 2 (single-source)	Face-swap	49.78%	0.525
Phase 6 (mixed-source)	StyleGAN	95.49%	0.995
Phase 6 (mixed-source)	Face-swap	82.42%	0.928

Interpretation

- On the face-swap test set, the mixed-source model achieves 82.4% accuracy and 0.928 AUC (vs. 49.8%/0.525 for Phase 2) — closing roughly 85% of the total possible gap, while StyleGAN performance drops only slightly (95.5% vs 98.7%).
- This is the thesis's strongest positive result up to this point: the generalization failure was primarily a data-diversity problem, not an architectural limitation.

10. Phase 7 — Third-Domain Generalization Test (Zero-Shot)

Both the Phase 2 and Phase 6 models were evaluated on Stable Diffusion-generated faces, a generation paradigm entirely unrepresented in either model's training data (a zero-shot test). Evaluation set: 2,536 fake / 5,000 real (imbalanced — AUC is the reliable metric).

Model	Accuracy*	AUC
Phase 2 (single-source)	65.5%	0.277
Phase 6 (mixed-source)	66.4%	0.552

Interpretation

- Phase 2's AUC (0.277) is below chance: it systematically rates diffusion fakes as more likely real than actual photos (99.8% miss rate on fakes).
- Phase 6's AUC (0.552) corrects this bias and is a real improvement, but remains far from reliable detection — mixed-source training's benefit only partially transfers, in a zero-shot setting, to a completely unrelated third forgery paradigm.

11. Phase 8 — Frequency-Domain Detection

Tests whether representing images via their 2D FFT log-magnitude spectrum, rather than raw RGB pixels, improves cross-domain generalization (SOTA Section 4.2). A ResNet18 was fully fine-tuned (all layers unfrozen) on frequency-domain StyleGAN images only, then evaluated on the same three domains used in Phases 2, 3, and 7.

Domain	Phase 2 (RGB) AUC	Phase 8 (Frequency) AUC	Change
StyleGAN (in-distribution)	0.999	0.920	-0.079
Face-swap (cross-dataset)	0.525	0.560	+0.035
Diffusion (third-domain)	0.277	0.303	+0.026

Interpretation

- In-distribution performance was noticeably lower than Phase 2 (0.920 vs. 0.999 AUC), and early overfitting was observed (best validation accuracy reached at epoch 3 of 5).
- Cross-domain AUC improved only marginally on both unseen domains (+0.035 face-swap, +0.026 diffusion) — directionally positive, but far smaller than Phase 6's improvement.
- Of the remedies tested up to this phase, training data diversity (Phase 6) remained substantially more effective than a change of input representation (Phase 8) alone.

12. Phase 9 — Three-Source Mixed Training (Few-Shot Recovery)

Phase 7 tested zero-shot generalization to diffusion-generated faces (no diffusion examples in training) and found only a partial benefit from two-source training. This phase tests the natural follow-up: does including even a modest sample of the third domain directly in training close the remaining gap? A ResNet18 was fine-tuned (identical recipe to Phases 2 and 6) on a three-source training set combining StyleGAN, face-swap, and 1,500 diffusion-generated images (out of 2,536 available; the remaining 1,015 were held out, disjoint from training, for evaluation, to avoid data leakage). All three models (Phase 2, Phase 6, Phase 9)

were then re-evaluated on this same held-out diffusion set, alongside fresh StyleGAN and face-swap test samples, for a fair three-way comparison.

Results:

Model	Domain	Accuracy	AUC
Phase 2 (single-source)	StyleGAN	98.70%	0.999
Phase 2 (single-source)	Face-swap	49.79%	0.523
Phase 2 (single-source)	Diffusion (held-out)	49.51%	0.285
Phase 6 (two-source)	StyleGAN	95.80%	0.995
Phase 6 (two-source)	Face-swap	82.54%	0.928
Phase 6 (two-source)	Diffusion (held-out)	53.65%	0.553
Phase 9 (three-source)	StyleGAN	96.01%	0.994
Phase 9 (three-source)	Face-swap	82.09%	0.913
Phase 9 (three-source)	Diffusion (held-out)	98.62%	0.999

Interpretation

- On the held-out diffusion domain, Phase 9 achieves 98.6% accuracy and 0.999 AUC — compared to Phase 2's 0.285 AUC (below chance) and Phase 6's 0.553 AUC (barely above chance). Including just 1,500 training images (roughly 2.4% of the 61,500-image three-source training set) took diffusion detection from catastrophic failure to near-perfect, a far larger effect than any zero-shot remedy tested in this work (Phases 4, 6-zero-shot, and 8).
- This improvement came at negligible cost to the other two domains: StyleGAN AUC changed by less than 0.001 (0.995 to 0.994), and face-swap AUC decreased only modestly (0.928 to 0.913), indicating the model gained substantial new capability without materially sacrificing previously learned capability.
- This result refines the conclusions of Phases 7-8 in an important way: the model's limitation with respect to diffusion-generated faces was not a fundamental inability to learn the relevant discriminative features, but specifically a zero-shot extrapolation limitation. Given even modest direct exposure to the target domain, the capacity to learn it was clearly present and readily realized.
- This constitutes the thesis's single strongest empirical result, and has a direct practical implication: a small, targeted data-collection effort for a new, previously unseen forgery type is likely to be dramatically more effective than any zero-shot architectural or representational remedy (ensembling, frequency-domain input) tested in this work.

13. Phase 10 — CLIP-Based Detector (Foundation Model Adapter)

Tests whether a frozen CLIP (ViT-B/32) vision backbone, pretrained on web-scale image-text data far broader than ImageNet, yields representations that generalize better zero-shot to unseen forgery types than an ImageNet-pretrained ResNet18 (SOTA review, Section 4.4). For a clean, isolated comparison, a small linear classification head (the only trainable component; all CLIP parameters remain frozen) was trained on CLIP embeddings of StyleGAN images only — identical training data to Phase 2 — then evaluated zero-shot on face-swap and diffusion-generated faces.

Results (AUC comparison against Phase 2's ResNet18):

Domain	Phase 2 (ResNet18) AUC	Phase 10 (CLIP) AUC	Change
StyleGAN (in-distribution)	0.999	0.988	-0.011
Face-swap (zero-shot)	0.525	0.484	-0.041
Diffusion (zero-shot)	0.277	0.232	-0.045

Interpretation

- In-distribution, CLIP's linear-probe performance (98.8% accuracy / 0.999 AUC for Phase 2 vs. 94.4% / 0.988 for CLIP) is slightly lower, consistent with CLIP's representation being fixed and general-purpose rather than specifically fine-tuned for this task.

- Critically, CLIP did not generalize better, zero-shot, to either unseen domain: its Face-swap AUC (0.484) and Diffusion AUC (0.232) were both at or below Phase 2's corresponding figures (0.525 and 0.277), with the Diffusion AUC in particular falling even further below the chance level of 0.5 than Phase 2's did.
- This result does not support the hypothesis that a broader-pretrained foundation-model backbone alone improves zero-shot generalization to unseen forgery types. Combined with the Phase 9 finding, this provides converging evidence that the dominant factor governing generalization in this work is specifically whether relevant training examples of a given forgery type were present during training — not backbone architecture, pretraining scale, or input representation. This rules out a plausible alternative explanation for the generalization gap (an insufficiently powerful or narrowly pretrained backbone) and correspondingly strengthens the practical conclusion drawn from Phase 9: targeted data collection, not architectural or representational substitution, is the most direct and effective remedy identified in this work.

14. Overall Conclusion (Phases 1-10)

Across ten phases, this work establishes a complete and nuanced empirical narrative regarding the generalization behavior of deepfake detectors. A fine-tuned ResNet18 achieves excellent in-distribution performance (98.8% accuracy, 0.999 AUC) on a single generator family (Phases 1-2), but this performance does not reflect a generally robust detector: the model collapses to near-chance (or below-chance) accuracy on unseen forgery types (Phase 3, Phase 7), and none of architectural ensembling (Phase 4), a change of input representation to the frequency domain (Phase 8), nor substituting a substantially broader-pretrained foundation-model backbone (Phase 10) meaningfully address this in a zero-shot setting. Diversifying the training data to include a second forgery type (Phase 6) substantially improves generalization to that type, though this benefit only partially extends, zero-shot, to a third, unrelated forgery paradigm (Phase 7). Critically, directly including even a modest, few-shot sample of that third domain in training (Phase 9) resolves the remaining gap almost entirely, at negligible cost to performance on the other domains. Taken together, these results converge on a specific, actionable conclusion: the generalization limitations observed throughout this work are attributable specifically to the absence of relevant training examples, rather than to any architectural, representational, or backbone-pretraining limitation of the underlying model. Adversarial robustness (Phase 5) remains an entirely separate, unaddressed weakness throughout, unrelated to the data-diversity findings that otherwise dominate this work's conclusions.

15. Next Steps

- Adversarial training (or other defense) to address the Phase 5 robustness gap, unaddressed by any phase to date.
- Given the strength of the Phase 9 finding and its reinforcement by Phase 10, investigate the minimum number of target-domain training examples required for substantial recovery (e.g. repeat Phase 9 with 100, 500, and 1,000 diffusion images instead of 1,500, to characterize the few-shot learning curve).
- Feature-level investigation of the Phase 7/8/10 below-chance AUC finding as a focused, self-contained sub-question — understanding specifically what visual property causes multiple independent models/backbones to systematically mistake diffusion-generated faces for real photographs.
- Fine-tuning (rather than linear-probing) the CLIP backbone itself, as a follow-up to Phase 10, to test whether allowing CLIP's representation to adapt (rather than remaining fully frozen) changes its generalization behavior.

Code & results (Colab notebooks, outputs included): [\[GitHub repository link\]](#)